

Kashtre Shared Unit Engine

Testing & navigation guide · Branch `demo` · For Inventory dual-run packaging

Share this with anyone verifying units, mappings, GRN conversion, and the Units console.

Important: This is not Kashtre super-admin home alone. Enter / browse a business first (e.g. *Exquisite Test Life*), then use Inventory screens.

1. Prerequisites

1. App running locally (e.g. `http://localhost:8000`).

2. In `.env` :

```
UNIT_ENGINE_ENABLED=true
```

Optional: `UNIT_ENGINE_STRICT=false` (keep false while dual-running; true = throw instead of silent legacy fallback)

3. Run once:

```
php artisan config:clear
php artisan units:install
```

4. Log in and open the target business (Inventory context).

2. Where things are

What	Where in the app	URL (typical)
Units console (catalog, mappings, rules, composites, governance, policies, audit)	Inventory sidebar → Units	<code>/inventory/units</code>
Item unit names dictionary + Map/Remap to core catalog	Sidebar → Manage Item Units	<code>/item-units</code>
Packaging on a product (sale unit, order unit, factor)	Manage Items → edit a <i>Good</i>	<code>/items/.../edit</code>
Receive goods (GRN) — sale units from pack qty	Inventory → Receive / Goods received	under <code>/inventory/...</code>
Orders / LPO prefill	Inventory → Orders / Purchase orders	under <code>/inventory/...</code>

Units console tabs

Tab	Purpose
Catalog	Browse SYSTEM + org units; create tenant packaging/count units
Mappings	PENDING / MAPPED / UNMATCHED Item Unit names → core units
Packaging rules	Per-item order→sale factors (created when you save packaged items)
Composites	Build ratio units (e.g. mg per mL)
Governance	Activate / deprecate / Verify UCUM (tenant units only)
Policies	Preferred units for CLINICAL / LIMS / INVENTORY objects
Audit	Recent governance & policy events

Item form fields (this is the real packaging config)

- **Sale unit** — stock / issue unit (e.g. Tab, Cap)
- **Order unit** — pack unit from supplier (e.g. Box) — must differ from sale unit for packaging to matter
- **Sale units per order unit** — factor (e.g. 100 tabs per box)

Manage Item Units only stores names (Box, Tab...). Reuse them from those dropdowns; do not create duplicate “Box” rows.

3. Recommended test path

Step A — Item units & mapping

1. Open `/item-units`.
2. Confirm **Box** (and Tab/Cap/Vial as needed) exist once per business.
3. Check **Engine mapping** column: Box should show `Mapped · BOX`.
4. For a custom name (e.g. Test Unit) still PENDING: click **Map** → pick a core unit → confirm.

Pass:

Badge shows `Mapped · CODE`. Creating a second “Box” for the same business is blocked.

Step B — Configure packaging on an item

1. Open **Manage Items** → edit a **Good**.
2. Sale unit ≠ Order unit (e.g. Cap/Tab vs Box).
3. Sale units per order unit = `100`.
4. Save (engine syncs public IDs + packaging rule).

Pass:

Item saved. Under Units → Packaging rules, a rule appears (e.g. box → tab × 100).

Step C — Receive goods (main check)

1. Inventory → create a GRN / receive goods.
2. Add that item: qty (boxes) = 3 , conversion = 100 .
3. Save.

Pass:

Sale units purchased = **300**. Stock increases by 300 SUOM when GRN posts.

Fail:

Sale units = 3 with factor 100 → same sale/order unit, unmapped names, or engine off.

Step D — Orders / LPO (optional)

- Order line with sale qty 300 → OUOM qty should be ~3.
- Prefill GRN from LPO with remaining SUOM 300 → delivery qty ~3, conversion 100.

Step E — Units console smoke

1. **Catalog** — search mg ; optionally create a tenant unit.
2. **Composites** — preview/save e.g. MG_PER_DL (mg / dL).
3. **Governance** — activate / verify a tenant unit; check **Audit**.
4. **Policies** — save a CLINICAL policy (analyte id + unit) if useful for demos.

4. Commands (ops)

Command	Use
<code>php artisan units:install</code>	Seed SYSTEM catalog (+ composites like MG_PER_ML)
<code>php artisan units:install --map-legacy --link-items</code>	Also map legacy names / link items
<code>php artisan units:dedupe-item-units</code>	Soft-delete duplicate Item Unit names per business
<code>php artisan units:dedupe-item-units --dry-run</code>	Preview only

5. Mental model (for the reader)

Item Units = name dictionary (Box, Tab).

Manage Items = which names + factor on each product.

Shared Unit Engine = canonical catalog + safe conversion (dual-run with legacy factor).

GRN / Orders = where conversion shows up in real stock/docs.

Symptom	Likely cause
Amber “engine off” on Units	<code>UNIT_ENGINE_ENABLED</code> false or config cached
Empty catalog	Run <code>units:install</code>
GRN 3→3 with factor 100	Sale = order unit, or unmapped units
Two “Box” rows	Duplicates — delete one; use <code>units:dedupe-item-units</code>
PENDING mapping	Map from Item Units or Units → Mappings

6. Code / config pointers (developers)

- Config: `config/units.php` · env `UNIT_ENGINE_ENABLED` , `UNIT_ENGINE_STRICT`
 - Gateway: `app/Domain/Units/Services/InventoryUnitGateway.php`
 - Console UI: `app/Livewire/Inventory/UnitEngineConsole.php`
 - Item Unit Map: `app/Livewire/ItemUnits/ListItemUnits.php`
 - Clinical / LIMS adapters: `ClinicalUnitGateway` , `LimsUnitGateway`
 - API (optional): `/api/v1/unit-engine/...`
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